



RB004 Week 1 — English Worksheet

Topic: Server vs Client — Who Owns the World? · **Course:** Arena Legends · **Time:** about 45 minutes

This week you learn the most important rule of multiplayer games: the **server** owns the world. Game scripts go in **ServerScriptService**, and each player's own computer is called a **client**. You test multiplayer with the **Test tab** → **2 Players**.

1 · Predict 🧠

Read each script. Before you imagine running it in Studio, write what you think will happen.

```
-- Script in ServerScriptService
local Players = game:GetService("Players")

Players.PlayerAdded:Connect(function(player)
    print(player.Name .. " joined the arena!")
end)
```

You press **Test** → **2 Players**. Player1 joins, then Player2 joins. What does the server print? How many lines?


```
-- Script in ServerScriptService
local gate = workspace.ArenaGate
gate.BrickColor = BrickColor.new("Bright red")
```

The server changes the gate's color. You test with 2 Players. Does Player1 see a red gate? Does Player2? Why?


```
-- LocalScript in StarterPlayerScripts
local player = game:GetService("Players").LocalPlayer

print("Hello, " .. player.Name .. "!")
```

A LocalScript runs on each player's own computer. With 2 Players, what does Player1's screen print? What does Player2's screen print?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The server is supposed to print a welcome message **every time** a player joins the arena.

```
-- Script in ServerScriptService
local Players = game:GetService("Players")

print(Players.LocalPlayer.Name .. " joined the arena!")
```

Hint: the server is not a player — it has no `LocalPlayer`. The server needs to **listen** for players joining.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This welcome script should see **every** player who joins. But it was put in the wrong place.

```
-- LocalScript in StarterPlayerScripts
local Players = game:GetService("Players")

Players.PlayerAdded:Connect(function(player)
    print(player.Name .. " joined the arena!")
end)
```

Hint: the code is fine — the **place** is wrong. A server script in the wrong place never sees everyone.

Write the fixed code (including where it goes):

Why was it wrong? Why does your fix work?

Bug C — The server should print the player's name plus the words " joined the arena!" as one message.

```
-- Script in ServerScriptService
local Players = game:GetService("Players")

Players.PlayerAdded:Connect(function(player)
    print(player.Name + " joined the arena!")
end)
```

Hint: in Luau,  is only for numbers. Strings join with a different symbol.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your own arena game in Roblox Studio. Add a server script in **ServerScriptService** that prints a welcome message when a player joins, then test it with **Test tab** → **2 Players**. When you finish, come back here.

Send a photo or video of the Output window showing both players' messages, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I put my script in **ServerScriptService** because ...

When I tested with 2 Players, I saw ...

The server is different from a client because ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you test your game with 2 Players. Talk through it like you are teaching someone who has never seen Roblox Studio. Try to use these words: **server, client, ServerScriptService, PlayerAdded, join**.

1. Show your script inside ServerScriptService, then press Test → 2 Players.
2. Point at the Output window and read what the server printed.
3. Say which things the **server** owns and which things belong to **one player only**.
4. Say in your own words why game scripts go on the server.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.